

GPU memory (constant)

$$\mathcal{L}_i = \{ {}^{i-1}\mathbf{r}_i, {}^{i-1}\mathbf{R}_i, {}^i\mathbf{r}_{c,i}, {}^i\mathbf{R}_{c,i}, m_i, \mathbf{I}_{c,i}, \text{diag}(\mathbf{a}) \}$$

$${}^{i-1}\mathbf{r}_i, {}^{i-1}\mathbf{R}_i$$

$${}^i\mathbf{r}_{c,i}, {}^i\mathbf{R}_{c,i}, \\ m_i, \mathbf{I}_{c,i}, \text{diag}(\mathbf{a})$$

GPU thread

$$\tau_{k,t}$$

Forward
Dynamics
Eq.(6)

$$\ddot{\mathbf{q}}_{k,t}$$

Eq.(7)

$$\mathbf{q}_{k,t}, \dot{\mathbf{q}}_{k,t}$$

Forward
Kinematics

$$\mathbf{J}_{k,t}, \dot{\mathbf{J}}_{k,t}$$

Compute
Dynamics

$$\mathbf{M}_{k,t}, \mathbf{h}_{k,t}$$

$$\mathbf{P}_{ee,k,t}$$